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Breast Cancer Detection from Whole-Slide Imaging Using Deep Learning

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Table of Contents

1. Introduction
2. Research Objectives
3. Literature Review / Related Work
4. Gap Analysis
5. Problem Statement
6. Research Questions
7. Methodology
8. Dataset & Sources
9. Evaluation Strategy & Metrics
10. Expected Outcome(s)
11. References

MS Thesis Proposal

1. Introduction

Breast cancer is one of the most common cancers among women. According to the World Health Organization (WHO), 2.3 million women were diagnosed with breast cancer and 670000 deaths all over the world in 2022 [1]. The disease originates in the cells of the breast tissue, with uncontrolled cell growth leading to the formation of malignant tumors. Various factors contribute to the development of breast cancer, including genetic predisposition, hormonal imbalances, age, and lifestyle choices. Early detection significantly improves patient prognosis, emphasizing the importance of advanced diagnostic techniques. Histopathological analysis remains a cornerstone of breast cancer diagnosis, playing an essential role in identifying tumor characteristics and guiding treatment decisions.

1.1. Background/Domain Concepts

Histopathological examination involves the microscopic evaluation of tissue samples obtained through biopsy procedures. These samples are typically stained with specialized dyes, such as Hematoxylin and Eosin (H&E) and Immunohistochemistry (IHC) markers, to highlight cellular structures and specific biomarkers. Pathologists carefully analyze these stained slides to identify cancerous cells, determine their characteristics, and assess the extent of tumor invasion. Critical biomarkers in breast cancer include Estrogen Receptor (ER), Progesterone Receptor (PR), and Human Epidermal Growth Factor Receptor 2 (HER2). The Allred scoring system is commonly employed to assess ER and PR status, combining the proportion of positive cells and staining intensity to yield a score ranging from 0 to 8. HER2 status is evaluated based on the intensity and completeness of membrane staining, with scores of 0 and 1 considered negative, 2 equivocal, and 3 positive [2]. Equivocal cases often require further testing, such as fluorescence in situ hybridization (FISH), to determine HER2 gene amplification [3]. However, the traditional manual approach to histopathological evaluation presents several challenges. The process is time-consuming, labor-intensive, and susceptible to subjective variability among pathologists.

1.2. The Problem

In recent years, artificial intelligence (AI) and deep learning approaches have emerged as promising tools to enhance the accuracy and efficiency of histopathological evaluations. Hasan et al. [2] combined segmentation models such as SegNet and U-Net with ResNet, VGG, and DenseNet classifiers to assist in biomarker scoring for ER and PR, achieving notable segmentation and classification F1 scores of 0.902 and 0.833, respectively. Similarly, Rehman et al. [3] employed a two-stage classification process using Vision Transformer (ViT) as the best-performing model to classify HER2 statuses from whole-slide images, achieving 99.9% patch-level accuracy and 78.8% Generalization accuracy. Goyal et al. [4] proposed a multi-model approach integrating CNNs, Vision Transformers, and clinicopathologic data to predict breast cancer recurrence risk as an alternative to costly genomic assays.

Despite their contributions, these methods remain fragmented, relying on separate models for different tasks. They do not comprehensively integrate HER2, ER, and PR biomarker analysis

alongside Allred scoring in a unified pipeline, limiting their scalability and clinical utility. Many traditional Artificial Intelligence techniques are used to work with whole slide images. As these are extremely large containing gigapixel images, but only a small portion of these images is annotated for histopathological features, this sparse annotation leads to challenges in building an accurate deep learning model. The size of these images also creates computational difficulties requiring capturing both local (cellular) and global (tissue-level) features. Also, existing models that use simple CNN-based models often focus on small-scale tasks e.g. analyzing specific patches or cells. Hence, there is a need for an automated AI platform that uses an advanced deep learning model that accurately processes the full whole slide images, and segments the region of interest followed by classification and scoring.

1.3. Motivation

The motivation for this research stems from the need to develop an integrated AI-driven framework that can automate the entire histopathological analysis process. By dividing whole-slide images into overlapping patches and employing multitask and multiresolution feature extraction, the proposed system aims to enhance the accuracy and efficiency of biomarker assessment. Implementing a multi-task learning module with dedicated heads for segmentation, classification, and scoring allows for shared learning across tasks, improving generalization and reducing computational costs. This comprehensive approach has the potential to significantly improve clinical decision-making in breast cancer care.

1.4. Scope

This research aims to develop an AI-driven framework for histopathological analysis of breast cancer, focusing on integrating biomarker evaluation and Allred scoring into a unified pipeline. By Focusing on advancements in AI-driven histopathological analysis for breast cancer diagnostics from 2023 to 2025, and excluding genomic-based detection methods, radiological imaging, and non-histopathological diagnostics, this study seeks to address current limitations and pave the way for more effective and scalable solutions that benefit pathologists, healthcare professionals, and patients alike.

2. Research Objectives

The primary objectives of this research are defined to address the limitations in AI-driven histopathological analysis for breast cancer diagnostics.

- Develop a unified pipeline for biomarker analysis and Allred scoring using Vision Transformers and multi-task learning approaches. This pipeline will streamline the assessment of HER2, ER, and PR biomarkers from whole-slide images, addressing the current fragmentation in diagnostic methods.
- Quantitatively assess the accuracy, efficiency, and scalability of the proposed solution through performance evaluation against traditional and state-of-the-art models. Key performance metrics, including classification accuracy, segmentation precision, and computational efficiency will be tracked.
- Ensure that the research outcomes directly contribute to improving breast cancer diagnostics by providing a comprehensive and automated solution for biomarker

assessment. This solution will aid pathologists in making more accurate and consistent clinical decisions.

- Complete the development, training, and evaluation of the proposed solution by 2025. Intermediate milestones will include model development, testing, and optimization phases to ensure timely progress and completion.

3. Literature Review

Advancements in deep learning have significantly contributed to the automation of histopathological analysis for breast cancer detection. Several studies have explored the use of convolutional neural networks (CNNs), Vision Transformers (ViTs), and hybrid models for biomarker assessment, segmentation, and classification in whole-slide imaging (WSI).

Hasan et al. [2] developed a real-time segmentation and classification model to assist pathologists in breast cancer diagnostics. Their study employed SegNet and U-Net architectures with modified decoders incorporating ResNet, VGG, and DenseNet for classification tasks. While achieving high segmentation (F1-score: 0.902) and classification performance (F1-score: 0.833), their approach did not consider HER2 biomarker analysis, restricting its clinical applicability. Furthermore, the reliance on convolutional architectures limits the model's ability to capture long-range dependencies, which are essential for holistic tissue analysis.

In a study by Rehman et al. [3], a two-stage classification framework was introduced for the classification of Normal, Amplified, and Non-Amplified HER2-SISH slide regions. Vision Transformer (ViT) was identified as the best-performing model, achieving a patch-level accuracy of 99.9%. However, the generalization accuracy dropped to 78.8% when applied to whole-slide images, highlighting the challenge of integrating local and global tissue information. Additionally, the study focused exclusively on HER2, neglecting the broader biomarker landscape essential for comprehensive breast cancer characterization.

Shamai et al. [6] attempted to predict ER, PR, and HER2 receptor status from H&E-stained slides using a hybrid deep learning approach. They employed ResNet-50 for feature extraction and a Transformer-based multiple-instance learning (MIL) framework for aggregation. While ER prediction demonstrated strong performance (AUC = 0.955), PR (AUC = 0.807) and HER2 (AUC = 0.846) exhibited lower accuracy, limiting the model’s ability to distinguish among breast cancer subtypes effectively.

Saranyaraj et al. [7] explored the classification of HER2 and ER biomarkers from histopathology images using stacked sparse autoencoders (SSAE) combined with deep CNNs. Their model achieved high classification accuracy (97.52%), however, it was computationally expensive and lacked external validation, raising concerns about real-world deployment. Additionally, their approach ignored PR biomarker evaluation, which is crucial for targeted therapies.

Systematic review on Vision Transformer applications in histopathology and cytopathology by Santos et al. [8] highlights the growing role of ViTs in medical imaging. By analyzing 73 studies, it emphasizes the importance of segmentation as a preprocessing step to improve classification accuracy and efficiency. Zhang et al. [9] introduced a multi-task deep learning model integrating CNNs and ViTs for tumor segmentation and classification in breast ultrasound images. Their architecture utilized parallel CNN and Transformer streams to capture both local and global

features, improving generalization on external datasets. Despite its promising performance, the model was designed for ultrasound images rather than histopathology slides. While these studies have contributed significantly to the field, they exhibit key limitations such as fragmented biomarker assessment, lack of multi-resolution learning, and the exclusion of a unified framework for segmentation, classification, and scoring. Addressing these gaps, our proposed methodology introduces a multi-task model that integrates HER2, ER, and PR biomarker evaluation alongside Allred scoring. By using multi-resolution feature extraction and joint segmentation-classification modules, our approach aims to enhance generalization, reduce computational costs, and improve clinical scalability.

Below is a detailed literature review table summarizing key studies relevant to breast pathology analysis, their methodologies, Results, and identified Limitations.

Author & Year	**Objective**	**Dataset**	**Model used**	**Methodology**	**Results**	**Limitation**	**Relevance**
Hasan et al. (2024)	Develop a real-time segmentation and classification model for breast cancer histopathology images to assist pathologists	ER-IHC stained private dataset & H&E-stained public MoNuSeg dataset	Utilized segmentation architectures, SegNet and U-Net, and modified the decoder to attach ResNet, VGG and DenseNet to perform classification tasks.	(1) Modified SegNet and U-Net decoders with ResNet, VGG, DenseNet backbones (2) Real-time processing with GUI interface	Segmentation F1-score: 0.902 (private), 0.903 (public), Classification F1-score: 0.833 (private)	(1) HER2 Not used in training or testing (2) Limited ability to model long-range dependencies (3) Lower classification accuracy highlights the need for advanced mechanisms like attention models for improvement	Provides insights into real-time segmentation, classification pipelines and architectural enhancements with decoders, though lacking HER2 analysis, a critical focus in our research.
Rehman et al. (2024)	Develop a deep-learning approach to classify Normal, Amplified, and Non-Amplifie-d regions in HER2-SISH slides	Private dataset of HER2-SISH breast cancer slides (46 WSIs, digitized at 40× magnificati-on)	Vision Transformer (ViT) (best model), DenseNet121, VGG16, MobileNetV2	Two-stage approach: (1)Patch-level classification on annotated regions (2) WSI-level regional classification with pseudo-color maps	Patch-level accuracy: 99.9%, Generalizati on accuracy: 78.8%,	Generalization accuracy lower than patch-level	Highlights ViT's potential for patch-based classification while emphasizing challenges in whole-slide analysis, reinforcing the importance of integrating both local and global tissue

							features, which we aim to address.
Che et al. (2023)	Develop an automated method for HER2 IHC scoring in whole-slide images (WSI) using deep learning	95 HER2 pathological WSIs from Peking Union Medical College Hospital (Sep–Dec 2021)	ResNet34	(1) Extracts tumor areas and segments WSIs into patches. (2) Uses ResNet34 for binary classification (tumor/non-tumor)	Patch-level precision: 95.77%, F1-score: 83.09%, Slide-level classification accuracy: 97.9%	Single biomarker (HER2) considered: Does not address ER and PR, limiting its clinical applicability.	Demonstrates HER2 scoring techniques and patch-based segmentation but lacks comprehensive biomarker evaluation, which we incorporate in our approach.
Shamai et al. (2024)	Predict ER, PR, and HER2 receptor status from H&E-stained slides using deep learning	(1) 19,845 H&E slides from 7,950 patients. (2)Multi-institutional dataset from six independent cohorts: TCGA, ABCTB, Carmel, Haemek, CHUCB, Ipatimup.	ResNet-50 for feature extraction, Transformer based Multiple Instance Learning (MIL). for aggregation.	(1) H&E slides segmented into non overlapping tiles. (2)Resnet-50 extracts tile level features (3)Transforme-r based MIL framework predicts receptor status. (4)Five fold cross validation with diverse datasets.	ER prediction: AUC: 0.955, PR prediction: AUC: 0.807, HER2 prediction: AUC 0.846	(1)Does not distinguish between ER-low and ER-high cases (2)The model performed well for ER but had lower accuracy for PR and HER2, which limits its ability to categorize breast cancer into subtypes.	Supports the need for hybrid models to improve biomarker prediction. Highlights the potential for Transformer-based aggregation techniques in histopathology.
Saranyaraj et al. (2023)	Early detection of breast cancer by classifying HER-2 and ER biomarkers from histopathology images.	653,400 microscopic image patches from 363 patients (BreakHis database)	Stacked Sparse Autoencoder (SSAE), deep Convolutional Neural Network (DCNN)	(1)Extracted epithelial regions from H&E-stained images (2)Applied SSAE for feature extraction (3)Classified HER-2 and ER using DCNN	Accuracy: 99.73% for nuclei classification, HER-2 and ER classification accuracy: 97.52%.	(1)High computational cost. (2)The study focuses only on HER-2 and ER biomarkers, ignoring other critical biomarkers like PR (3)Lack of External Validation	Demonstrates effective classification but highlights computational inefficiency and limited biomarker scope, reinforcing the need for a balanced computational approach in our research.

Zhang et al. (2024)	Develop a multi-task deep learning model for tumor segmentation and classification in breast ultrasound images.	(1)-BUSI-SPH (1959 images: 1238 benign, 721 malignant) (2)-GDPH&SYSUCC (2405 images: 886 benign, 1519 malignant, used for external validation)	CNN + Transformer (ViT)	(1) Parallel CNN & Transformer streams to extract local and global features. (2) Multi tumoral region guidance module for classification. (3) Soft parameter sharing for segmentation and classification tasks.	(1) The model outperforms state-of-the-art methods in tumor segmentatio-n and classificatio n. (2)Demonstr-ates strong generalizatio-n on external datasets.	Fixed-size peri-tumoral mask may not adapt well to varying tumor sizes.	Highlights multi-task learning benefits and CNN-Transfor mer synergy, offering valuable insights for our multi-resolutio -n feature extraction and classification model design.

4. Gap Analysis

Existing AI-based approaches for breast pathology analysis exhibit several limitations. CNN-based models fail to capture long-range dependencies essential for holistic tissue-level analysis [2]. Most studies focus on individual biomarkers, neglecting comprehensive biomarker evaluation critical for accurate diagnosis and personalized treatment [5]. Additionally, the lack of a unified framework for segmentation, classification, and Allred scoring limits the clinical applicability of these models. Moreover, manual analysis of WSIs by pathologists is time-consuming, subjective, and prone to inter-observer variability, particularly during Allred scoring.

To overcome the challenges in breast pathology analysis and limitations of existing AI techniques, we propose a solution that divides whole-slide images into overlapping patches to capture boundary details effectively. Our approach utilizes Vision Transformers combined with CNNs and attention mechanisms to extract multi-resolution feature maps, capturing both cell-level (local) and tissue-level (global) information. A multi-task learning framework is employed, featuring specialized heads for segmentation, classification, and scoring. The segmentation head identifies regions of interest, the classification head detects molecular markers, and the scoring head, through a fully connected network, predicts the Allred score using regression.

5. Problem Statement

Current deep learning methods for breast cancer analysis struggle with the vast size of WSIs and sparsely annotated regions, leading to loss of tissue-level information [2]. Additionally, most methods lack an integrated framework for segmentation, classification, and scoring of ER, PR, and HER2 biomarkers, limiting clinical scalability and effectiveness [5]. So, the problem at hand is to develop a multiresolution and multi-task learning model that can make use of both local(cell) and global(tissue) level features to solve the segmentation, classification, and scoring under a single model.

6. Research Questions

Primary Research Question:

- How can a multi-resolution and multi-task deep learning model be developed to effectively segment, classify, and score ER, PR, and HER2 biomarkers in breast cancer whole slide images (WSIs) using both cell-level and tissue-level features?

Secondary Research Questions:

- How can overlapping patch-based processing improve boundary detail capture in WSIs for more accurate segmentation and classification?
- What role does the integration of CNNs and Vision Transformers play in extracting comprehensive multi-resolution features for breast pathology analysis?
- How does the proposed multi-task learning framework enhance the simultaneous segmentation, classification, and scoring of biomarkers compared to existing single-task approaches?

7. Methodology

In recent years, advanced deep learning techniques have emerged as powerful tools for breast pathology analysis, addressing challenges in the segmentation and classification of histopathology images. One notable study by Hasan et al. [2] employed a hybrid approach combining segmentation architectures such as SegNet and U-Net with classification models including ResNet, VGG, and DenseNet. Additionally, supported by a graphical user interface for seamless user interaction. The models were trained and tested on both private (ER-IHC-stained) and public (H&E-stained MoNuSeg) datasets, demonstrating reliability across different magnifications (20x and 40x). Despite its strengths, the approach faced certain limitations. HER2, a crucial biomarker for breast cancer analysis, was not used during the training or testing process, potentially affecting the model's diagnostic utility. Additionally, the method exhibited limited ability to model long-range dependencies, which are vital for capturing global contextual information. The lower classification accuracy highlighted the need for advanced mechanisms, such as attention-based models, to improve performance. Moreover, the system's computational complexity posed challenges for resource-constrained environments. To address these issues, the integration of attention-based mechanisms could further enhance the system's ability to capture subtle variations, ultimately improving diagnostic precision.

Another study by Zhang et al. [9] adopted a novel approach by combining Vision Transformers (ViTs) and convolutional neural networks (CNNs) in a multi-task learning framework. This

architecture simultaneously addressed segmentation and classification tasks, showing superior performance compared to using separate models for each task. The combination of CNN's feature extraction capabilities and the global attention mechanism of Vision Transformers allowed the model to capture both fine-grained and contextual information effectively. This approach was specifically applied to ultrasound images, demonstrating significant performance improvements over existing methods. Building on its success, this approach can be extended to whole slide images for breast pathology analysis, to further enhance diagnostic accuracy and efficiency.

To address the problem in breast pathology analysis and limitations with the previous AI techniques we proposed a solution that divides the whole slide images into smaller patches for processing. To capture the boundary details patches are overlapping. Then for segmentation, classification, and scoring an advanced deep learning model such as Vision Transformers will use a convolutional neural network and attention mechanism to get the multi-resolution features map that captures both local(cell) and global(tissue-level) details then these features passed to a multitask learning module that makes use of multiple heads, one head for each task. A segmentation head segments the region of interest and a classification head classifies the molecular markers using CNN and average pooling layers, at last a fully connected network is used as a scoring head that predicts the Allred score using a regression. This approach will not only automate the whole process but also improve the accuracy of segmentation and classification tasks as multi-resolution techniques allow the model to capture both fine-grained cellular detail and broader tissue-level patterns. On the other hand, the multitask learning approach enables the model to perform multiple tasks at a time by sharing learned features between them, it improves the generalization and reduces the computational costs

8. Dataset & Sources

This research acknowledges the availability of two publicly accessible datasets, the Post-NAT-BRCA and the Early Breast Cancer Core-Needle Biopsy (BCNB) datasets, alongside a proprietary dataset from the Pakistan Institute of Medical Sciences (PIMS).

Post-NAT-BRCA Dataset:

- **Source:** Accessible through The Cancer Imaging Archive (TCIA) at https://www.cancerimagingarchive.net/collection/post-nat-brca/.
- **Structure and Attributes:** This dataset comprises 96 Hematoxylin and Eosin (H&E) stained whole slide images (WSIs) from 54 patients with breast cancer. Each WSI is accompanied by clinical features and expert pathology annotations, including tumor cellularity assessments and cell type classifications. The images are stored in uncompressed .svs file format, scanned at 20x magnification using an Aperio slide scanner. Additionally an Excel (.xlsx) file contains clinical features for each patient, such as age, treatment details, and ER/PR/HER2 status. Each row in the spreadsheet corresponds to a single (.svs) slide.
- **Ethical Considerations:** The dataset has been anonymized to protect patient identities, with all personal identifiers removed. Its use complies with ethical standards for research involving human subjects.
- **Justification for Dataset Selection:** The Post-NAT-BRCA dataset provides high-resolution WSIs accompanied by detailed clinical and pathological annotations,

including ER, PR, and HER2 status. This comprehensive information is essential for developing and validating deep learning models aimed at assessing residual tumor cellularity and other pathological features in breast cancer patients.

BCNB Dataset:

- **Source:** Available through the Grand Challenge platform at https://bcnb.grand-challenge.org/.
- **Structure and Attributes:** The BCNB dataset includes 1,058 H&E stained WSIs from early breast cancer patients. Each image has been examined and annotated by two independent and experienced pathologists. The dataset also provides comprehensive clinical data for each patient, encompassing age, tumor size, tumor type, ER, PR, HER2 status, histological grading, molecular subtype, number of lymph node metastases, and the metastatic status of axillary lymph nodes (ALN). The WSIs are provided in .jpg format, with slides scanned using an Iscan Coreo pathologic scanner at 200x magnification.
- **Ethical Considerations:** The BCNB dataset has been desensitized, with all personal identifiers removed, ensuring compliance with ethical standards for research involving human subjects.
- **Justification for Dataset Selection:** The BCNB dataset is particularly relevant for this research as it provides a substantial number of WSIs accompanied by detailed clinical and pathological annotations, including ER, PR, and HER2 status. The availability of clinical data for these biomarkers allows for effective validation of the models' predictive capabilities.

PIMS Dataset:

While the Post-NAT-BRCA and BCNB datasets offer valuable resources, they have limitations in terms of detailed pathological annotations specific to certain biomarkers. To address these gaps, this research will incorporate breast pathology slides from the Pakistan Institute of Medical Sciences (PIMS) Pathology Department. Expert histopathologists will perform segmentation and provide comprehensive annotations, including detailed pathological features and classifications.

- **Source:** Breast pathology slides from the PIMS Pathology Department.
- **Justification for Dataset Selection:** Utilizing the PIMS dataset addresses the limitations of the publicly available datasets by providing detailed pathological annotations tailored to the specific requirements of this research. This approach ensures a comprehensive dataset that enhances the depth and applicability of the study.

Preprocessing Steps:

To prepare the datasets for analysis, the following preprocessing steps will be performed:

1. **Normalization:** Stain normalization techniques will be applied to reduce variability in staining intensity and color, thereby enhancing the performance of the model.
2. **Segmentation:** Tissue segmentation will be implemented to isolate regions of interest,

such as tumor areas, from the background. This step involves distinguishing between relevant tissue structures and non-informative regions, facilitating focused analysis on pertinent areas.

3. **Patch Extraction:** Large whole slide images (WSIs) will be divided into smaller, manageable patches to facilitate training and improve computational efficiency. This approach allows for detailed analysis of high-resolution images without overwhelming computational resources.
4. **Data Augmentation:** Transformations such as rotation, flipping, and scaling will be applied to increase the diversity of the training data and prevent overfitting. By augmenting the dataset, the model can generalize better to unseen data, improving its predictive performance. Advanced techniques, including Generative Adversarial Networks (GANs), may also be employed to generate synthetic images, further enhancing data diversity.

By integrating these datasets and applying preprocessing steps, this research aims to develop an effective deep learning model for breast cancer pathology, using the strengths of each data source to address the research problem comprehensively. These preprocessing steps are essential to standardize the data, focus the analysis on relevant regions, and enhance the model's ability to generalize, thereby ensuring reliable and accurate outcomes.

9. Evaluation Strategy & Metrics

To evaluate the effectiveness of the proposed multi-resolution and multi-task deep learning approach for breast pathology analysis, the following evaluation strategy and metrics will be employed:

Performance Metrics:

- **Segmentation:** Intersection over Union (IoU) and Dice Coefficient to evaluate the accuracy of segmenting regions of interest.
- **Classification:** Accuracy, Precision, Recall, and F1-score for molecular marker detection.
- **Scoring:** Root Mean Squared Error (RMSE) to assess the Allred score prediction accuracy.

These metrics are selected based on their effectiveness in measuring the segmentation accuracy, classification efficiency, and regression performance, which are critical for breast pathology analysis.

Comparative Analysis:
The proposed model will be compared with state-of-the-art models, including traditional CNN-based approaches and transformer-only models, to demonstrate performance improvements.

Experimental Setup:

- **Data Splitting:** A train-test-validation split will be used to ensure effective evaluation.

- **Validation Techniques:** Cross-validation will be employed to mitigate overfitting and assess model generalizability.

10. Expected Outcomes

The proposed solution is expected to yield the following outcomes:

- **Improved Segmentation and Classification Accuracy:**
  The model will accurately segment regions of interest and classify molecular markers by capturing multi-resolution features, improving diagnostic reliability.
- **Enhanced Allred Scoring Precision:**
  A fully connected network with a regression head will deliver precise Allred scores, reducing inter-observer variability.
- **Reduced Computational Costs:**
  The multitask learning approach will enable simultaneous segmentation, classification, and scoring, leading to efficient resource usage and reduced computation time.
- **Holistic Pathology Analysis:**
  By capturing both cell-level and tissue-level details, the model will provide a comprehensive pathology analysis, enhancing clinical decision-making.
- **Contribution to the Field:**
  This research will set the foundation for scalable AI solutions in breast pathology, potentially transforming clinical workflows and improving diagnostic accuracy in real-world healthcare settings.

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